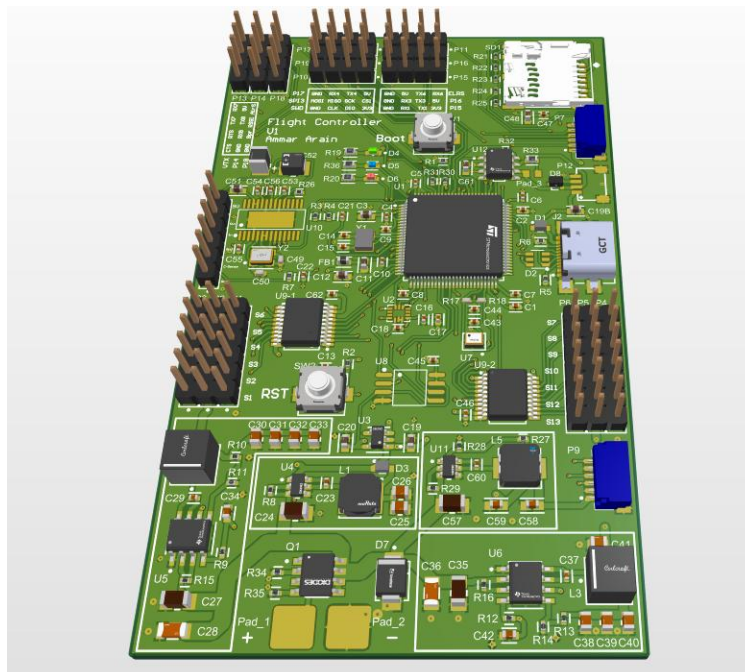

Flight Controller H7

V1

By Ammar Arain



Overview

This flight controller runs natively on the stock MatekH743 ArduPilot target, with no custom code compilation needed. The battery voltage sensor uses a 10k:100k voltage divider and the battery pads offer **some** overvoltage and reverse polarity protection (though the MOSFET might get damaged and could potentially need to be replaced, but the main board will remain safe). The board also has a SPI flash memory chip on the same SPI pins as the IMU and CS (chip select pin) connected to PE2 on the STM32H7. This remains unused by default on ArduPilot, but is available if needed on custom firmware. The air speed (AirS), RSSI and current sensor header pins only accept a maximum of 3.3V, if the sensor exceeds that you will need to add an external voltage divider.

Components

MCU: STM32H743VIT6, 480MHz, 2MB Flash

IMU: ICM-42688-P

Barometer: DPS368XTSA1

Analog OSD: AT7456E

Blackbox: MicroSD card slot

Peripherals

7x UARTs (1,2,3,4,6,7,8)

13x PWM outputs

1x I2C

1x CAN

4x ADC (VBAT, Current, RSSI, AirSpeed)

3x LEDs for Status (red, blue) and 3.3V indicator (green)

1x SPI3 header pins

USB Type-C (USB 2.0)

2x JST-SH_6pin (Digital VTX: Rx7, Tx7, Rx8 – GPS: Rx2, Tx2, I2C1)

1x JST-SH_4pin for CAN

Since the pinout is the same as MATEK H743 V4 Slim, the timer groupings are identical as well. Below is a screenshot from MATEK's [website](#):

PWM PWM1~PWM13 are Dshot and PWM capable. However, mixing Dshot and normal PWM operation for outputs is restricted into groups, ie. enabling Dshot for an output in a group requires that ALL outputs in that group be configured and used as Dshot, rather than PWM outputs. If servo and motor are mixed in same group, make sure this group run lowest PWM frequency according to the servo specification. ie. Servo supports Max. 50Hz, ESC must run at 50Hz in this group.	S1	PB0	5 V tolerant I/O	PWM1 GPIO50	TIM8_CH2N	Group1
	S2	PB1	3.3 V tolerant I/O	PWM2 GPIO51	TIM8_CH3N	
	S3	PA0	5 V tolerant I/O	PWM3 GPIO52	TIM5_CH1	Group2
	S4	PA1	5 V tolerant I/O	PWM4 GPIO53	TIM5_CH2	
	S5	PA2	5 V tolerant I/O	PWM5 GPIO54	TIM5_CH3	
	S6	PA3	5 V tolerant I/O	PWM6 GPIO55	TIM5_CH4	
	S7	PD12	5 V tolerant I/O	PWM7 GPIO56	TIM4_CH1	Group3
	S8	PD13	5 V tolerant I/O	PWM8 GPIO57	TIM4_CH2	
	S9	PD14	5 V tolerant I/O	PWM9 GPIO58	TIM4_CH3	
	S10	PD15	5 V tolerant I/O	PWM10 GPIO59	TIM4_CH4	
	S11	PE5	5 V tolerant I/O	PWM11 GPIO60	TIM15_CH1	Group4
	S12	PE6	5 V tolerant I/O	PWM12 GPIO61	TIM15_CH2	
	S13/LED	PA8	5 V tolerant I/O	PWM13 GPIO62	TIM1_CH1	Group5
					SERVO13_FUNCTION 120, NTF_LED_TYPES neopixel	

Power

9V for VTX

Input: 2-6S LiPo

Current Sensor: Requires external sensor, connects to header pin

Rail 1 (S1 to S6): 5V 3A

Rail 2 (S7 to S13): 5V 3A

Component/Module Power (connects with LDO & UART header pins): 5V 2A

LDO: 3.3V 600mA

VTX Power: 9V 2A

FC Firmware

ArduPilot: MATEKH743

BetaFlight: MATEKH743

INAV: MATEKH743

Physical

Dimensions with JLCPCB Edge Rails: 71mm x 99mm

Dimensions without Edge Rails: 61mm x 99mm

Weight: 32g (with Edge Rails)

To maximize component density and keep the board footprint as small as possible, this layout does not use traditional 30.5mm stand-off holes.

Recommended Mounting: The board is designed to be soft-mounted flat against the airframe or inside a 3D-printed case using double-sided foam tape. This is a common practice for fixed-wing/VTOL builds and provides superior vibration isolation for the IMU compared to hard nylon standoffs.

Flight Controller pins (shows JLCPCB edge rails on left and right side):

